

WYVERN-E — Research Proposal

A Skylight Rocketry Venture · 70 mm Single-Stage Servo-TVC Sustainer

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Executive Summary

WYVERN-E is a 70 mm, single-stage, **3D-printed active-thrust-vector-control (TVC)** research rocket that demonstrates closed-loop flight stabilization on a bare-metal **Raspberry Pi Pico 2 W (RP2350)** flight computer, powered by the **Estes F15-4**. It carries four 72 mm fins (no ballast) for **passive stability during launch and the F15 ignition spike**, after which ($t \geq 0.5$ s) the TVC loop engages on the smooth portion of the thrust curve to stabilize and execute a commanded maneuver. The two candidate TVC actuators — a tri-solenoid *magnetic* gimbal and a *servo* gimbal — are compared quantitatively on a purpose-built **3-axis thrust-vector balance** that resolves both thrust magnitude and vector direction; the flight vehicle carries the servo system. The program retains the static motor/materials test regimen; the aerofoil wind-tunnel campaign was removed in 2026-08 and its aerodynamic question is now answered analytically and validated in flight (RQ3). All flights are FAA Class-1 (no waiver, no certification). Predicted apogee ≈ 429 ft; project cost $\approx \$1,725$.

1. Background & Motivation

1.1 Thrust vector control

Thrust vectoring steers a rocket by gimbaling the exhaust, producing a control moment without aerodynamic surfaces — the method used by every orbital launch vehicle. A small, low-cost, fully printed TVC vehicle makes the control problem tangible while producing real flight data.

1.2 WYVERN lineage

WYVERN-E 1.0 (interceptor study) → 2.0 (84 mm two-stage, custom avionics) → 3.0 (Raspberry Pi 5, solenoid-vs-servo A/B in flight) established the aero, structures, and avionics groundwork. 4.0 distills those lessons into the simplest vehicle that still answers the core question: a single stage, a single bare-metal controller, and the A/B actuator comparison moved to a repeatable ground balance.

1.3 The 4.0 thesis

A small finned rocket can use passive stability to survive launch and the ignition transient, then hand authority to a closed-loop TVC system on the smooth thrust curve — and the relative merit of magnetic vs servo gimbals can be measured directly on the ground before flight.

2. Research Questions

Numbering matches WYVERN_E4_Proposal_rev2.md §2 (canonical). The wind tunnel and the airfoil-CFD package were removed from the program in 2026-08; the aerofoil-polar question they served is replaced by RQ3 below, which is answered analytically and validated in flight.

#	Question	Method A	Method B	Primary metric
RQ1	Magnetic vs servo TVC actuation	3-axis thrust-vector balance, F15-0	bench signal model + SIL	bandwidth, slew, overshoot, SSE, max vector angle
RQ2	Zoned print-material structure + jetvane erosion	3-point bend coupons; F15-0 plume on static stand + deflector	lumped-capacitance thermal + first-order FEA	flexural stiffness, mass-loss, char depth, HDT margin
RQ3	Predicted vs in-situ passive stability	Barrowman CP/CN + RK4 dispersion	flight-telemetry reconstruction of margin and coast Cd	Δ static margin (cal), Δ Cd
RQ4	Closed-loop gain sensitivity in flight	24-point phase/gain-margin sweep + Monte Carlo	onboard log, up to 4 flights	pitch error, gimbal track, PM/GM, recovery

3. Vehicle Architecture

3.1 Configuration & mass budget

70 mm OD, 0.74 m, three bays + two bulkheads; 4 fins (72 mm), no ballast.

Section	Key mass	Subtotal
Nose (ASA-Aero)	ellipsoid nose	21 g
Recovery bay (ASA-Aero)	bay tube, chute+Kevlar cord, Nomex, BNO085 (vote), PC-FR bypass tube, PC-FR sealed Bulkhead B, ejection plenum	137 g
FC bay (ASA-Aero)	bay tube, Pico 2 W, BNO085, baro, μ SD, i3 4K Thumb Action Camera, 2S LiPo + 5 V UBEC	122 g
Engine/TVC bay (PC-FR)	bay tube, Bulkhead A,	268 g

	gimbal, 2 servos, BNO085, mount	
Structure	4 ASA-Aero fins (72 mm) + wiring	50 g
Dry		603 g
Motor	F15-4 (60 g prop)	102 g
Liftoff		705 g

3.2 Materials

- **PC-FR** (ρ 1.25 g/cm³, HDT ~110 °C, UL94-V0): both bulkheads, ejection bypass tube, and the engine assembly (engine/TVC bay + motor mount + gimbal) — heat/ejection-pressure zones only.
- **ABS** (ρ ~1.04 g/cm³, HDT ~98 °C, not flame-rated): **jetvane material coupon only** (exhaust erosion screen alongside ASA-Aero); not a structural airframe material.
- **ASA-Aero** (foamed, ρ ~0.65 g/cm³): nose, body tube, fins, FC/recovery bays — primary structure, saves ~130 g where there is no motor heat. Thermal check: engine-bay wall peaks **~38 °C** over the 3.45 s burn (no liner needed; first-order lumped model, we4_analysis.py).

3.3 Stability — fins + ballast + the 0.5 s rule

An apogee sweep shows ballast lowers altitude, so we use **no ballast** and size fins to the minimum stable 1.0 cal: 4×72 mm fins — CP 56.8 cm, CG 49.1 cm = **+1.10 cal** static margin (stable). This passive margin holds the vehicle through launch and the F15 ignition spike; the TVC controller is **inhibited until t = 0.5 s**, then engages on the smooth curve. A historical finless variant (margin -5.6 cal) was rejected because it is statically unstable and cannot survive the pre-TVC transient.

3.4 Structural & thermal margins

First-order analysis (we4_analysis.py): minimum safety factor **> 300×** (the 25 N motor leaves the 1.6 mm airframe print/handling-limited, not load-limited); fin flutter velocity well above the 25 m/s flight regime; engine-bay thermal margin to PC-FR HDT > 70 °C.

4. Propulsion & Trajectory

4.1 Motors (verified)

Motor	Spec	Role	Qty
Estes F15-4	49.6 N·s, 14.4 N avg / 25.3 N pk, 3.45 s, 4 s delay + ejection	flight only (ejection = recovery)	4
Estes F15-0	49.6 N·s, 14.4 N avg / 25.3 N pk, 3.45	ground stands + jetvane	13

	s, 0-delay (plugged)		
Estes E16-4	~ 16 N avg	stand	6
		commissioning	

4.2 Predicted performance (unified RK4 + Barrowman, `we4_flightsim.py`)

T/W 2.08 avg / 3.66 peak; Cd 0.54; burnout 3.45 s, ~75 m, ~36 m/s; **apogee ~429 ft @ 6.82 s**; deploy forced t = 4.0 s @ ~29 m/s; 18" chute — ~6 m/s descent. Dispersion ($\pm 5\%$ mass, $\pm 15\%$ Cd): see plots4/06_dispersion.png.

5. Flight Computer & Control

5.1 Raspberry Pi Pico 2 W (RP2350)

A single dual-core 150 MHz RP2350 is flight computer *and* 500 Hz controller: core 0 reads three BNO085 + two baros (BME688 + BMP388) and closes the TVC loop driving 2 servos via hardware PWM; core 1 handles microSD logging and Wi-Fi bench telemetry. No Linux, no scheduler jitter. Recovery is the motor's own ejection charge (the FC only logs/observes).

5.2 IMUs — Game Rotation Vector

All three BNO085 run in **Game Rotation Vector** (accel + gyro, magnetometer disabled) to reject the magnetic interference of the adjacent servos. Gimbal deflection = $q_{body}^{-1} \otimes q_{gimbal}$, giving true nozzle attitude relative to the body (catches linkage backlash/flex). The recovery-bay unit votes against the FC unit for fault detection.

5.3 Control law

Per-axis PID ($K_p=0.10$, $K_i=0.40$, $K_d=0.18$; margin- and auto-tune-validated), output clamped to $\pm 8^\circ$, servo lag $\tau \approx 0.04$ s. **TVC inhibited for the first 0.5 s** (ignition spike) — fins hold attitude — then engages: stabilize to vertical, then a 4° commanded maneuver. Required gimbal torque 0.56 kg·cm (micro-servo class); control authority positive throughout the powered phase.

5.4 Power & data

A light 2S LiPo (7.4 V, ~450 mAh) feeds a single 5 V/6 V UBEC (set 5 V) whose one rail powers Pico 2 W VSYS, the camera, and both servos (EMAX ES08MA II, running at 5 V ≈ 1.8 kg·cm). Shared-rail decoupling (1000 μ F bulk at the servos, 100 μ F + an SS34 hold-up Schottky at VSYS) keeps servo-stall transients from browning-out the Pico. No recovery battery — recovery is the motor's own ejection charge. Pack voltage is monitored on GP26/ADC0 (100k/62k divider; warn 6.4 V, arm-inhibit 6.0 V). Onboard log: full-rate IMU/baro/control + i3 4K Thumb Action Camera 1080p60 video. The power+camera group (LiPo ~30 g + UBEC ~10 g + i3 4K Thumb Action Camera ~36 g ≈ 76 g) sits inside the 122 g FC-bay budget.

6. Recovery

Recovery uses the **F15-4 motor's own ejection charge** (4 s delay → fires $t \approx 7.45$ s, ~ 0.66 s past apogee), routed through a solid-walled 12 mm PC-FR bypass tube past the *sealed* FC bay into the recovery bay to release a friction-fit nose — **no RRC3+, no 9 V, no e-match/BP, no CO2, no FC involvement**. Feasibility (we4_ejection_feasibility.py): tube loss ≈ 0.06 kPa; bay pressurizes to ~ 140 kPa vs a 14–41 kPa nose-release threshold = **3.4× margin**. F15-4 is the closest Estes delay to the ~ 3.5 s coast optimum (F15-6/-8 eject 2.5 s/4.5 s late, too fast/low). Single passive event, no electronic backup. Opening at ~ 6.1 m/s; 1/8" Kevlar cord ($> 800\times$ margin) + Nomex protector; 18" chute → ~ 6 m/s.

7. Ground Test Program

7.1 3-axis thrust-vector balance

Motor + gimbal on a thrust block restrained by one axial (5 kg) + two lateral (1 kg) strain-gauge load cells → T, θ, ϕ . Actuator-agnostic — both magnetic and servo systems tested identically. RQ3 metrics logged vs commanded. ### 7.2 Static thrust + materials stand Axial cell + steel deflector: validates the F15-0 curve and screens jetvane materials in the plume (plugged 0-delay on the ground — no ejection into the fixture). ### 7.3 Motor & calibration plan Load cells dead-weight calibrated, then commissioned with **6 × E16-4** (3 per stand). Counts: F15-4 $\times 4$ (flight), F15-0 $\times 13$ (ground), E16-4 $\times 6$.

8. Simulation Suite (RQ3/RQ4 Method A)

RK4 + Barrowman trajectory with componentwise drag buildup, power-law wind shear, and Monte Carlo dispersion over the atmospheric and build-tolerance envelope; 500 Hz closed-loop pitch model with servo lag, transport delay, and Dryden-spectrum gust forcing; phase/gain-margin sweep across the burn-time $\times$ atmosphere grid; and a software-in-the-loop flight computer writing logs in the onboard recorder's schema. See Simulations/README.md.

9. Safety & Regulatory

Single F15-4: 49.6 N·s, 60 g propellant, $\leq$ F class, liftoff 705 g < 1500 g → **FAA Class-1, no waiver, no Level-1 certification**. Remote ignition, ≥ 3 m standoff on the stands, gimbal-neutral fail-safe, motor-integral ejection (igniter installed at the pad; no electronic ejection circuit to arm or inhibit).

10. Budget

$\approx$ **\$1,725** total program spend (vehicle + 3-axis balance + static/materials stand + one-time tools + all motors): \$1,246 still to buy + \$479 already acquired. This is down from the \$1,882 originally scoped; the Hofferth wind tunnel section was deleted from the BOM in the 2026-08 scope change. Live per-line pricing in Documentation/WYVERN_E4_BOM.xlsx. Per-flight consumable $\approx$ F15-4 \$17 (integral delay/ejection; no separate initiator or BP charge).

11. Schedule & Milestones

Phase

Print + assemble

Bench bring-up

TVC balance A/B (RQ1)

Static fires + jetvane (RQ2)

Sim + margin analysis (RQ3/RQ4)

Flight tests (RQ3/RQ4)

Analysis + paper

12. Risk Register

Risk	Likelihood	Mitigation
Pre-TVC instability	Med	4× 72 mm fins, no ballast (+1.10 cal at liftoff); TVC inhibit 0.5 s
Servo slew too slow	Med	bench-verify on balance before flight; fast digital micro
Hard deploy at ~6.1 m/s (motor ejection, +0.63 s past apogee)	Low	Kevlar harness >800× margin, Nomex protector, ground-tested charge
Launch-detect miss at low T/W	Low	tune arming alt; verify on 2.2 g spike
Camera/SD throughput	Low	self-contained i3 4K Thumb Action Camera (decoupled from FC)

13. Expected Outcomes & Deliverables

A flight-validated small TVC vehicle; a quantitative magnetic-vs-servo TVC dataset; a quantified accuracy bound on Barrowman-class stability prediction versus flight telemetry; jetvane material rankings; full open documentation (CAD, firmware, wiring, sims) and a research paper.

References

Barrowman, J. *The Practical Calculation of the Aerodynamic Characteristics of Slender Finned Vehicles*, NASA, 1967. Box, J. (BPS.space) *Signal/Echo* TVC flight series. NAR/Tripoli Model Rocket Safety Codes. Bosch Sensortec BNO085 datasheet; Raspberry Pi RP2350 / Pico 2 W datasheet.